Portland State UNIVERSITY

Critical Terms in Information Systems Ch. 16,17,18, and Emerging Technology

Adapted from Information Systems: A Manager's Guide to Harnessing Technology (v 10.1) by John Gallaugher

Chapter 16: Software in Flux

- Marginal costs
- Open source software (OSS)
- Cloud computing
- Software as a service (SaaS)
- Virtualization
- Scalability
- Utility computing
- Service level agreement (SLA)
- Vertical markets (vertical niches)
- Platform as a service (PaaS)
- Infrastructure as a service (IaaS)
- Server farm

Chapter 17: Data and Competitive Advantage

- Big Data
- Business Intelligence (BI)
- Analytics
- Machine learning
- Dynamic pricing
- Data
- Information
- Knowledge
- Database
- Database management system (DBMS)
- Structured query language (SQL)
- Database administrator (DBA)
- Table or file
- Column or field

- Row or record
- Relational database
- Serverless computing
- transaction processing systems (TPS)
- transaction
- loyalty program
- artificial intelligence (AI)
- data aggregators
- legacy system
- data warehouse
- data mart
- online analytical processing (OLAP)
- data cube
- data lake
- Hadoop
- Data cloud
- ETL
- Query tools
- Python
- R
- Graphical query tools
- canned reports
- ad hoc reporting tools
- dashboards
- data visualization
- omnichannel
- data mining
- over-engineer

Chapter 18: Artificial Intelligence

- Generative AI
- Prompts
- AI hallucination
- LLM
- Parameters (as part of training models)
- Corpus
- Artificial Intelligence
- Neural networks
- Machine learning
- Deep learning
- Foundation model
- Turing test

- Expert systems (as a category of AI systems)
- Genetic algorithms
- Supervised learning
- Self-supervised learning
- Reinforcement learning from human feedback
- Constitutional AI
- Pair programming
- Prompt injection
- Data poisoning
- Deepfakes
- OCR
- Change management

Emerging Technology

Blockchain:
When you look up each term, include the word ‘blockchain’ in your search to ensure you get definitions in the context of distributed-ledger technology. **Example:** Blockchain Decentralization

- Blockchain
- Block
- Ledger
- Hash
- Node
- Decentralization
- Consensus
- Smart Contract
- Cryptocurrency
- Bitcoin
- Ethereum
- Mining
- Wallet
- Public Key
- Private Key
- Gas Fee

Quantum Computing:

- Quantum computing
- Qubit
- Superposition
- Entanglement
- Quantum Gate
- Quantum Algorithm
- Quantum Supremacy
- Shor’s Algorithm

- Quantum Key Distribution (QKD)